

Project -02 | Application Trend

Step 01 – Clone the repo to local EC2 Ubuntu machine

git clone <https://github.com/Vennilavanguvi/Trend.git>

Step 01 – Write the Dockerfile to build image.

Vi Dockerfile

```
root@ip-172-31-35-67:/home/ubuntu/Trend# cat Dockerfile
```

```
FROM public.ecr.aws/nginx/nginx:alpine
```

```
COPY dist/ /usr/share/nginx/html
```

```
EXPOSE 80
```

```
CMD ["nginx", "-g", "daemon off;"]
```

Build the image locally to test it.

docker build -t trend-aj:v2 .

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
524781515606.dkr.ecr.us-west-2.amazonaws.com/brain-tasks-repo:v1	70736e4d65d2	93MB	26.1MB	U
jenkins/jenkins:lts	23b84d1066db	802MB	288MB	U
mindtrace-aj:v1	70736e4d65d2	93MB	26.1MB	U
moby/buildkit:buildx-stable-1	5601811fde88	333MB	104MB	U
myacradgrt.azurecr.io/python-app:1.2.0	28091f65421a	686MB	173MB	
singajmihub/python-app:1.2.0	28091f65421a	686MB	173MB	
trend-aj:v2	040b45416061	111MB	35MB	U
your-dockerhub-username/brain-tasks-dev:dev-1	b62904ecd28a	93MB	26.1MB	U
your-dockerhub-username/brain-tasks-dev:dev-2	c5b31b633d15	93MB	26.1MB	U

Step 03 – Setting Terraform to deploy IOC.

Setup the communication, authentication & authorization to AWS from ubuntu machine.

```
root@ip-172-31-35-67:/home/ubuntu/Trend# aws configure list
NAME      : VALUE                                : TYPE      : LOCATION
profile   : <not set>                               : None      : None
access_key : *****4Z2D                             : iam-role   :
secret_key : *****0j0t                             : iam-role   :
region    : us-west-2                               : imds       :
root@ip-172-31-35-67:/home/ubuntu/Trend# |
```

This enables to push the terraform code from local to AWS API.

Version Control -

Write the main.tf which containers VPC, IAM, EC2, main.tf code file is pushed to github repo - <https://github.com/singh-ajmer-git/trend.git>

Version control- Created a repo on Github and pushed the code in trend-infra branch.

Added the repo on local, initiated committed and pushed the code to repo. This includes giving permission to the report in PAT token.

```
root@ip-172-31-35-67:/home/ubuntu/Trend# git checkout -b dev
Switched to a new branch 'dev'
root@ip-172-31-35-67:/home/ubuntu/Trend# git branch
* dev
  main
root@ip-172-31-35-67:/home/ubuntu/Trend# git init
Reinitialized existing Git repository in /home/ubuntu/Trend/.git/
root@ip-172-31-35-67:/home/ubuntu/Trend# git add .
root@ip-172-31-35-67:/home/ubuntu/Trend# git commit -m "Terraform code to git"
[dev e5daff8] Terraform code to git
Committer: root <root@ip-172-31-35-67.us-west-2.compute.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

7 files changed, 472 insertions(+)
create mode 100644 .gitignore
create mode 100644 .terraform.lock.hcl
create mode 100644 .terraform/providers/registry.terraform.io/hashicorp/aws/6.35.1/linux_amd64/LICENSE.txt
create mode 100755 .terraform/providers/registry.terraform.io/hashicorp/aws/6.35.1/linux_amd64/terraform-provider-aws_v6.35.1_x5
create mode 100644 Dockerfile
create mode 100644 main.tf
create mode 100644 provider.tf
root@ip-172-31-35-67:/home/ubuntu/Trend#
```

Run terraform commands to push the code to AWS API

Terraform init

Terraform fmt

Terraform validate

Terraform plan

Terraform apply -auto-approve

Terraform state list

List of resources that running in AWS based on main.tf

```

root@ip-172-31-35-67:/home/ubuntu/Trend# terraform state list
aws_iam_instance_profile.ec2_profile
aws_iam_role.ec2_role
aws_instance.web
aws_subnet.trend_public_subnet
aws_vpc.trend_vpc
root@ip-172-31-35-67:/home/ubuntu/Trend#

```

Instances (3) [Info](#)



Find Instance by attribute or tag (case-sensitive)

All states

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
☐	ubuntu_guvi	i-051f93fe6ee412e5	Running	t3.small	3/3 checks passed	View alarms	us-west-2a
☐	jenkins	i-0f8829451ba6dc51c	Stopped	t2.micro	-	View alarms	us-west-2b
☐	trend-ec2	i-0fbb6154d058a50d4	Running	t2.micro	2/2 checks passed	View alarms	us-west-2a

Your VPCs

[VPCs](#) | [VPC encryption controls](#)

Your VPCs (3) [Info](#)

Last updated less than a minute

Find VPCs by attribute or tag

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR
☐	MY-VPN2	vpc-0ca583ade2698127f	Available	-	-	Off	10.0.0.0/16
☐	trend-by-aj	vpc-0099605c47d36e125	Available	-	-	Off	10.0.0.0/16
☐	VPN-lb	vpc-020409da3e274321b	Available	-	-	Off	172.31.0.0/16

[Details](#) | [Status and alarms](#) | [Monitoring](#) | [Security](#) | [Networking](#) | [Storage](#) | [Tags](#)

Security details

IAM role

[trend-ec2-role](#)

Owner ID

524781515606

Launch time

Sat Mar 07 2026 23:33:58 GMT+0530 (India)

Security groups

[sg-03620bdaf4a4fe4b4 \(default\)](#)

Install Jenkins & verify <http://54.244.138.14:8080/> user-singajm

Instance details | EC2 | x

VPC | us-west-2 | x

Instances | EC2 | us-we... | x

Group 1 | GUVI | x

Dashboard - Jenkins | x

Application Deploy... | x

singh-ajmer-git/trend | x

←

↻

⚠ Not secure

35.92.109.179:8080

New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up builds or start building a software project.

Start building your software project

Create a job

Set up a distributed build

Kubernetes Setup - Created Cluster by applying

```
eksctl create cluster -f cluster.yaml
```

```
aws eks update-kubeconfig --region us-west-2 --name trend-eks
```

Code pushed to Git-hub

<https://github.com/singh-ajmer-git/trend.git> branch trend-infra

Cluster is running fine on AWS

Clusters (1) Info							 Delete
Filter clusters							
	Cluster name	Status	Kubernetes version	Support period	Upgrade policy	Created	
☐	trend-eks	 Active	1.34 Upgrade now	 Standard support until December 2, 2026	Extended support	2 hours ago	

Following resources have been deployed via Jenkins pipeline

Local machine to Git HIB >> Jenkins >> trigger pipeline based on webhooks >> build image as instructed in Dockerfile >> deploy it on EKS cluster ..

Deployment.yaml

Service.yaml

Namespace – Monitoring

Prometheus + Grafana for monitoring

Now Define the Pipeline in Jenkins to run the CI/CD:

		trend-app-build	1 day 10 hr 	1 day 11 hr 	9.2 sec	
---	---	-----------------	---	---	---------	---



Jenkins / trend-app-build

Status

Changes

Workspace

Build Now

Configure

Delete Project

GitHub Hook Log

Rename

trend-app-build

trend-app-build-test

Permalinks

- [Last build \(#8\), 9 min 51 sec ago](#)
- [Last stable build \(#8\), 9 min 51 sec ago](#)
- [Last successful build \(#8\), 9 min 51 sec ago](#)
- [Last failed build \(#7\), 14 min ago](#)
- [Last unsuccessful build \(#7\), 14 min ago](#)
- [Last completed build \(#8\), 9 min 51 sec ago](#)

Builds >

Filter

Today

#9 6:34 PM

#8 6:23 PM



Jenkins / trend-app-build / Configure

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Connect and manage your code repository to automatically pull the latest code for your builds.

None

Git

Repositories

Repository URL

https://github.com/singh-ajmer-git/trend.git

Credentials

- none -

+ Add

Advanced

```
root@ip-172-31-30-99: /home X + v
root@ip-172-31-30-99: /home/ubuntu/trend/trend-infra# kubectl get node
NAME                                STATUS    ROLES    AGE    VERSION
ip-172-31-21-43.us-west-2.compute.internal Ready    <none>    135m   v1.34.4-eks-f69f56f
ip-172-31-35-167.us-west-2.compute.internal Ready    <none>    135m   v1.34.4-eks-f69f56f
root@ip-172-31-30-99: /home/ubuntu/trend/trend-infra# kubectl get deployment
NAME    READY    UP-TO-DATE    AVAILABLE    AGE
trend-app  2/2      2              2            33m
root@ip-172-31-30-99: /home/ubuntu/trend/trend-infra# kubectl get service
NAME    TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
kubernetes ClusterIP    10.100.0.1     <none>          443/TCP    141m
trend-app-service LoadBalancer 10.100.226.196 a5f54f9bc4adf4b8dbcef762783db15e-1449293500.us-west-2.elb.amazonaws.com 3001:31014/TCP 33m
root@ip-172-31-30-99: /home/ubuntu/trend/trend-infra# kubectl get pods
NAME    READY    STATUS    RESTARTS    AGE
trend-app-7f4867c99-5blff  1/1      Running    0            33m
trend-app-7f4867c99-vthzr  1/1      Running    0            33m
root@ip-172-31-30-99: /home/ubuntu/trend/trend-infra# kubectl get pods -n monitoring
NAME    READY    STATUS    RESTARTS    AGE
alertmanager-monitoring-kube-prometheus-alertmanager-0 2/2      Running    0            28m
monitoring-grafana-7b4d968bf4-hst6n 3/3      Running    0            5m36s
monitoring-kube-prometheus-operator-68cd9f9577-tpjjh 1/1      Running    0            28m
monitoring-kube-state-metrics-85c77b54f-qn6pc 1/1      Running    0            28m
monitoring-prometheus-node-exporter-4v22z 1/1      Running    0            28m
monitoring-prometheus-node-exporter-thw96 1/1      Running    0            28m
prometheus-monitoring-kube-prometheus-prometheus-0 2/2      Running    0            28m
root@ip-172-31-30-99: /home/ubuntu/trend/trend-infra#
```

← ↻ ⚠ Not secure aa6e67b92ed5f460aa2e1a815ac67f6a-1020500526.us-west-2.elb.amazonaws.com/?orgId=1&from=now-6h&to=now&timezone=browser

Grafana

×

Dashboards > Home

Home

Bookmarks

Starred

Dashboards

Explore

Drilldown

Alerting

Connections

Administration

Welcome to Grafana

Basic

The steps below will guide you to quickly finish setting up your Grafana installation.

TUTORIAL

DATA SOURCE AND DASHBOARDS

Grafana fundamentals

Set up and understand Grafana if you have no prior experience. This tutorial guides you through the entire process and covers the "Data source" and "Dashboards" steps to the right.

← ↻ ⚠ Not secure 44.255.63.67:9090/graph?g0.expr=&g0.tab=1&g0.display_mode=lines&g0.show_exemplars=0&g0.range_input=1h

Prometheus Alerts Graph Status ▾ Help

☐ Use local time

☐ Enable query history

☒ Enable autocomplete

☒ Enable highlighting

☒ Enable linter

🔍

Expression (press Shift+Enter for newlines)

Table

Graph

<

Evaluation time

>

Webhooks

Webhooks

Add webhook

Webhooks allow external services to be notified when certain events happen. When the specified events happen, we'll send a POST request to each of the URLs you provide. Learn more in our [Webhooks Guide](#).

● <http://44.255.112.51:8080/github-w...> (push)

Edit

Delete

This hook has never been triggered.

Finally the web is running in a container which is hosted on Kubernetes.

